

RUDRA CHAVDA

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EDUCATION

University of California, Santa Barbara
Bachelor of Science, Computer Engineering

Expected June 2027
GPA: 3.95

- **Honors:** 5x Dean's List
- **Coursework:** Data Structures and Algorithms, C Programming, Computer Architecture, Operating Systems, Software Engineering.

EXPERIENCE

AIEA Lab (UCSC)

Dec 2025 – Present

Undergraduate Researcher (Robustifying Autonomous Vehicles)

- Benchmarking Reinforcement Learning (RL) algorithms using **Stable Baselines** to evaluate autonomous vehicle safety.
- Deploying the **CARLA** simulator in a headless environment on the **Nautilus** hyper-cluster (Kubernetes).
- Analyzing edge-case failures by visualizing agent behaviors and reward functions post-training.

CA-Futures (UCSC EEB & Baskin Engineering)

May 2025 – Sep 2025

Software Developer

- Developed a full-stack conservation modeling application using **Vue.js** (frontend) and **Py4Web** (backend).
- Integrated geospatial analysis tools (**GDAL**, Ecoscape) to track species connectivity and visualize ecological data.
- Optimized simulation parameters to process large-scale biological datasets for the Ecology Lab.

Opennote (YC S25)

Jul 2025 – Present

Campus Growth Lead

- Spearheaded campus growth strategy, increasing user engagement by 30% and acquiring 50+ active users.

Fassur

June 2023 – Oct 2023

Front-End Engineering Intern

- Architected a secure, interactive web application using **React.js** and **Tailwind CSS**.
- Designed a responsive front-end framework to improve data visualization for client dashboards.

PROJECTS

Better Notes | *TypeScript, Tailwind CSS, Vercel*

Spring 2025

- Engineered a full-stack note-taking platform with a custom markdown rendering engine for student study workflows.
- Secured 100+ active users within 3 months of deployment to Vercel.
- Live Demo: betternotes-two.vercel.app

Div Explorer Project | *Machine Learning, Python*

Fall 2025

- Developed actionable data labeling pipelines to detect duplicate discoveries and high-overlap patterns.
- Implemented categorization metrics (interpretability, coverage, anomalies) to improve ML model fairness.

Personal Portfolio | *Next.js, Three.js, GSAP*

Winter 2024

- Built a high-performance portfolio site featuring 3D interactive elements using Three.js and GSAP animations.

LEADERSHIP

Eagle Scout Project

Summer 2024

Project Lead, Troop 537

- Directed a team of 20+ Scouts in the construction of redwood structures for Sycamore Grove Park (Livermore).

TECHNICAL SKILLS

Languages	Python, C, C++, Java, TypeScript, JavaScript, SQL, Assembly
Frameworks	Vue.js, React, Next.js, Py4Web, Tailwind CSS, Angular
Tools	Git, Nautilus (Cluster), CARLA, Docker, GDAL, VS Code, Google Colab
Concepts	Reinforcement Learning, Data Structures, Algorithms, System Design